

Figure 2.11 Wool Fiber

of dead animals. The wool obtained by shearing is called fleece or clipped wool. The wool obtained from dead animals is called pulled wool. Sometimes, chemicals are added in the food of animals which result in the loss of hair. This is wool is weak when compared to mechanically sheared wool. The pulled wool and sheared wool can be mixed for yarn formation.

2.4.2 Manufacturing of Wool

Wool Grading and Sorting

Wool fleece is classified based on the quality. It is graded based on the parts of animal from which it is sheared.

Best quality wool used for clothing- sheared from shoulder and sides.

Lesser quality wool used for rug making- sheared from lower legs.

Wool Cleaning and Scouring

The sheared wool has some natural oil present in them. This collected wool is known as grease wool. It is packed in bales then washed and sorted again. The recycled wool fibre is obtained by separately reducing the unused and used material to a fibrous mass by picking and shredding process called garneting. Further to remove the oil content, the fibres are washed in dilute acids, which is known as carbonizing. The resultant wool is called extract. Further the wool is washed with alkaline solution known as scouring. The washed wool is dried. Care is taken to leave 12 to 16% of moisture in the wool. The scoured wool is fluffy therefore it is treated with vegetable oil which helps in making the softness of wool suitable for spinning. The wool is further dyed and blended if required.

2.4.3 Characteristic of Wool Fibre

Shape	Fibre contain scales
Length	3.75
Elongation	Good
Fibre luster	Medium
Elasticity	More
Stretchability	More
Flexibility	More
Density	1.30 – 1.32 g/cm^3
Reaction with acid	Not affected
Reaction with alkalies	Most of alkalies destroy the fibre
Absorbency	Good
Reaction to sunlight	More heat affects the fibre.
Flame	Burns slowly in direct flame, but self extinguishing.
Insects	Damaged by moths and carpet beetles
Drying	Prolonged drying
Dyeing	Absorbs acid and basic dyes readily.

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